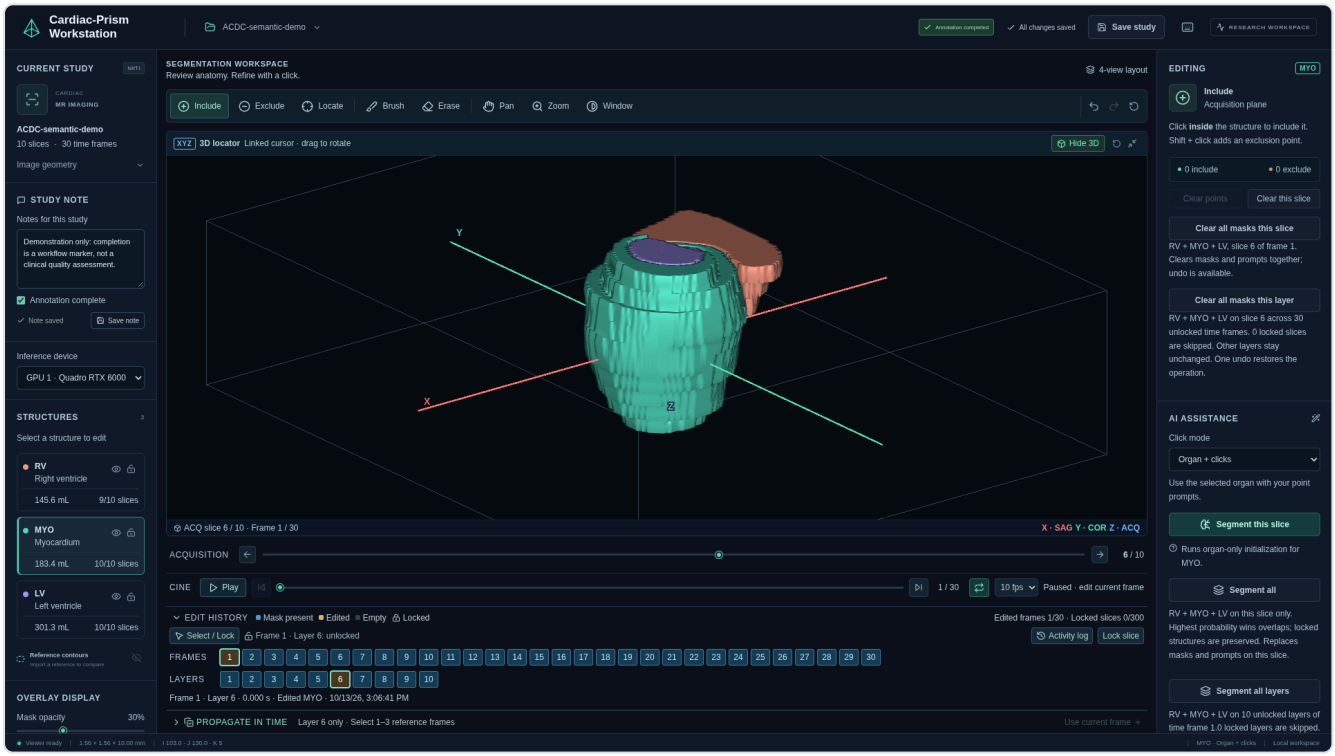
VIEW · REFINE · PRESERVE

Cardiac-Prism Workstation

Open, segment and review a study

Connect, open a scan, learn the segmentation tools, then refine and save your work.



Public ACDC cine with actual Cardiac-Prism predictions. This guide shows generated masks and illustrated edits, not ACDC expert reference masks.

Open and navigate
Connect, import and explore
Pages 2–5

Segmentation
Automatic, clicks and references
Pages 6–10

Review and preserve
Refine, play, lock and save
Pages 11–17

Server installation instructions are maintained in the GitHub repository. This guide focuses on using the workstation: connect on page 2, open a case and learn the controls, then follow the segmentation examples.

Guide prepared for the current 0.1.x workstation. Research software for noncommercial use; predictions require review. The public ACDC images are demonstration data, not a performance benchmark.

01 · OPEN THE WORKSTATION

Connect and open the workstation

> **Download and installation:** visit github.com/bangwayne/cardiac-prism. The repository provides the source, model weights and full server setup instructions. Once the service is running, follow the quick connection steps below.

Use Chrome or Edge on your Mac or Windows computer. Your administrator provides the server address and starts the workstation service.

1. **Open a terminal on your own computer.**
   On a Mac: open **Terminal** from Applications → Utilities.
   On Windows: open **PowerShell** or **Windows Terminal**.
2. **Run the command below.** Replace `YOUR_USERNAME` and `YOUR_SERVER` with the details your administrator provides. Use your own server account.

```
ssh -N -L 8502:127.0.0.1:8502 YOUR_USERNAME@YOUR_SERVER
```

Complete the normal login and any institutional authentication. Password characters may not appear as you type. After login, a quiet terminal is normal: it is keeping the connection open.

3. **Leave that terminal open.**
4. **Open Chrome or Edge** and type this into the address bar:

```
http://localhost:8502/?library=1
```

> **Run the tunnel on your own computer**, not inside an existing remote SSH session. Keep the terminal open while using the workstation.

If port 8502 is already in use on your computer

Use a different *local* port, while keeping the server port at 8502:

```
ssh -N -L 8512:127.0.0.1:8502 YOUR_USERNAME@YOUR_SERVER

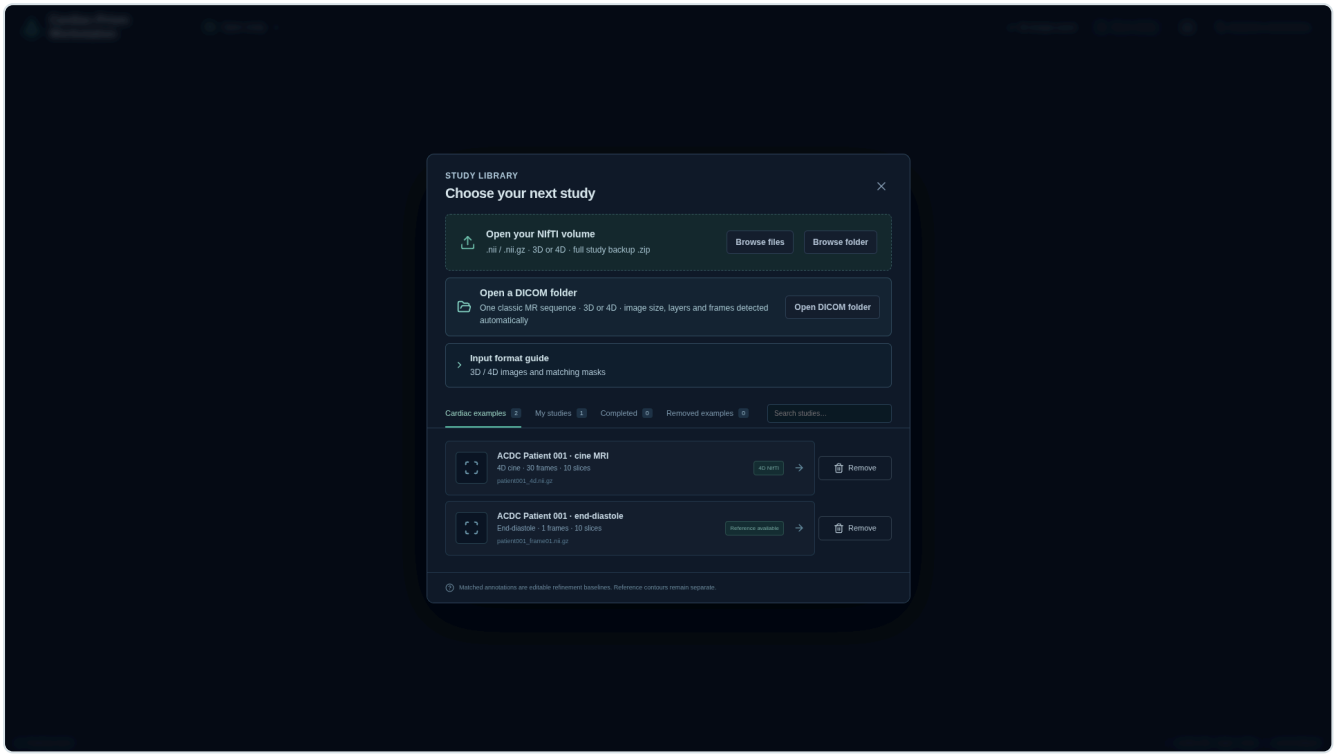
http://localhost:8512/?library=1
```

The browser address must match the first port in `-L`. Use 8512 only with the second command. `localhost` and `127.0.0.1` both refer to your own computer here.

You may also forward server port 8502 in your editor and open its reported local address. On Windows, enable OpenSSH Client if `ssh` is unavailable. Ctrl+C closes the tunnel.

02 · CHOOSE A STUDY

Begin with an example or your own image



The Library organizes inputs and saved working studies. Example availability depends on the server installation.

Library entry	What it does
Example card	Opens the example image as a working study, or reopens its existing study.
Open your NIfTI volume	Imports a `.nii` or `.nii.gz` image, or restores a workstation **Full study backup** ZIP.
Browse folder	Lists NIfTI files in a folder on your computer so you can choose an image.
Open DICOM folder	Imports one supported MR series from a folder. See page 4.
My studies / Completed	Returns to saved work. Search can match a study name or its notes.

> **An example is an input; a study is your working copy.** Your masks, clicks, notes and locks belong to the study. Opening an example does not turn the original image into your annotation storage.

ACDC cine: the supplied public example contains 30 time frames and 10 layers. The separate end-diastolic 3D example has an expert reference annotation. The original cine does not include a reference mask for every time frame.

Use **Library** in the workstation header whenever you want to return here. Save pending annotation changes before switching studies.

03 · INPUT FORMATS

Keep the original image size and geometry

Image	File dimensions	Example
3D NIfTI	`(X, Y, Z)`	`192 × 320 × 12` One volume with 12 layers.
4D NIfTI cine	`(X, Y, Z, T)`	`192 × 320 × 12 × 30` 30 volumes, each with 12 layers.

Width and height can differ and do not have to be 256 or 512. The model prepares its input internally and returns masks on the original grid. Use a scalar image: one intensity value per voxel. In 4D files, the last axis must represent time.

Choose **Image geometry** to inspect dimensions, spacing and input checks. Window and level affect only the display. Different acquisition protocols can still produce different prediction quality after intensity normalization.

A simple DICOM folder

1. Put **one uncompressed classic MR series** in a folder. Files can end in `.dcm` or have no extension.
2. Choose **Open DICOM folder**, select the folder, and accept the browser's folder-upload prompt.
3. Wait for the spatial/time checks and conversion. The study opens when ready.

This release supports complete 3D/4D frame–layer grids. It does not support enhanced multi-frame DICOM, compressed transfer syntaxes, DICOM ZIP files, mixed-series projects or PACS. Limits: 1,000 DICOM files, 512 MiB total, 128 million voxels. NIfTI uploads are limited to 512 MiB and 128 million voxels.

Bring an existing annotation

Open the image first. In **Import annotation**, choose **Editable annotation** to replace unlocked working masks, or **Reference comparison** to add a comparison overlay. Then choose the NIfTI annotation file and its label format.

0 · Background **1 · RV** **2 · MYO** **3 · LV**

For one structure, choose **Binary mask → RV / MYO / LV**. The annotation must match image dimensions, spacing, origin and direction; a 4D image needs a matching 4D annotation. The importer does not silently resample misaligned masks.

> **A 3D ACDC reference belongs with its matching 3D phase image.** Do not attach that file directly to the full 4D cine.

04 · THE LINKED VIEWS

One position, one time frame



Actual model predictions on the ACDC cine. All views share one position and time cursor; annotation takes place in the acquisition plane.

Move through the scan

Locate moves the shared crosshair. Use the layer controls or ↑ / ↓ to change acquisition layer. View titles describe scan-grid directions; an oblique scan may not align with anatomical planes.

Pan, **Zoom** and **Window** change the view. **Fit** returns the image to a comfortable size. The corner expansion control enlarges one view while preserving the shared position.

Choose what you see

Select **RV**, **MYO** or **LV** in the structure list before editing. Eye buttons show or hide structures; they do not lock them.

Mask opacity and **Contours only** help reveal the underlying image. **Locator color / opacity** adjust the crosshair. **Point opacity** hides markers without deleting clicks.

> **Edit in the acquisition plane.** Clicks, brush strokes and erasing affect the selected organ at the current frame and acquisition layer. The other views help you inspect that location in context.

Default colors: RV coral, MYO green, LV purple. NIfTI and DICOM use the same palette. The 3D locator initially shows axes; use **Render 3D** to create mask surfaces.

05 · THE LEARNING SEQUENCE

Three ways to create a mask

Start with an organ name, start with clicks, or give the model an example image and mask.

1 · Semantic segmentation

Input: the image and a selected organ, such as MYO.
Action: choose **Segment this slice**.
Result: an initial prediction without placing any points.

Useful when you want the model to identify a named structure automatically.

2 · Interactive segmentation

Input: the image and positive/negative points.
Action: begin with an empty mask, choose **Clicks only**, then use **Include** and **Exclude**.
Result: a mask generated from your spatial prompts.

Start fresh with **Clicks only**, or use **Organ + clicks** to refine a semantic prediction with the organ name and spatial prompts together.

3 · In-context segmentation

Input: a reference image with its accepted editable mask, and a target image.
Action: select reference frames in **PROPAGATE IN TIME**.
Result: predictions on other time frames at the same layer.

A target can be empty. For a difficult basal or apical layer, use a carefully reviewed reference frame on that same layer to guide other time frames.

All three methods remain slice-based. One prediction addresses a time frame and acquisition layer. Whole-frame and whole-cine buttons repeat this processing over the requested positions.

Understand the two sources of waiting

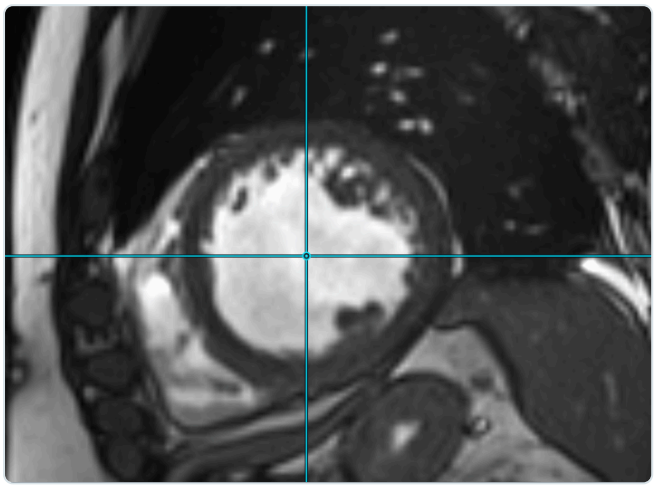
Inference runs on the server device selected in the UI. **Drawing the views** runs in the browser. Model loading, computation, transfer and rendering contribute separately to the time you see.

The segmentation demonstrations use an NVIDIA A100 server GPU on Achilleas. The recording browser uses software rendering, which is separate from server inference. Initial model loading, transfer and drawing can still add delay; other installations may have different timings.

06 · PREDICT FROM THE ORGAN NAME

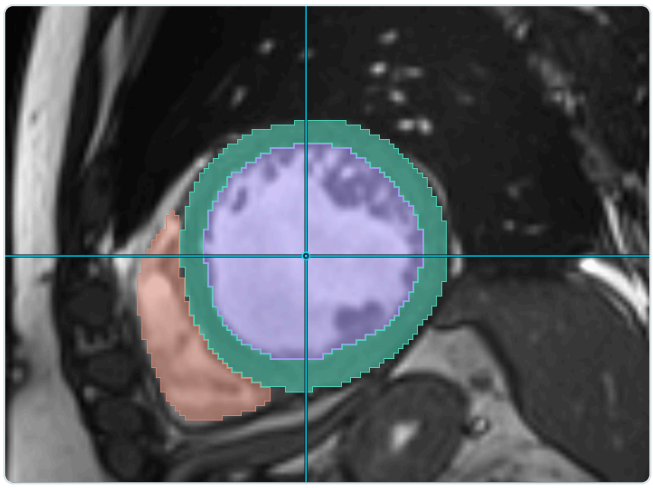
Create the first mask without clicks

Before · empty working mask



Frame 1, Layer 6. The image is present; no mask or point prompt has been supplied.

After · semantic prediction



Segment all predicts RV, MYO and LV in the illustrated example.

1. Pause and select the time frame and acquisition layer.
2. Choose **RV**, **MYO** or **LV** in the structure list.
3. Click **Segment this slice** for one organ, or **Segment all** for all three, as illustrated. No point is needed.
4. Wait for processing, inspect the boundary, and refine it if necessary.

Button	Processing scope
Segment this slice	Selected organ, current frame and layer.
Segment all	RV, MYO and LV, current frame and layer.
Segment all layers	All three organs, unlocked layers of the current time frame.
Segment all time frames	All three organs, unlocked frame/layer positions throughout the cine.

All-structure segmentation assigns competing editable labels using the highest predicted probability. Lock accepted slices before a larger job if those annotations should be retained.

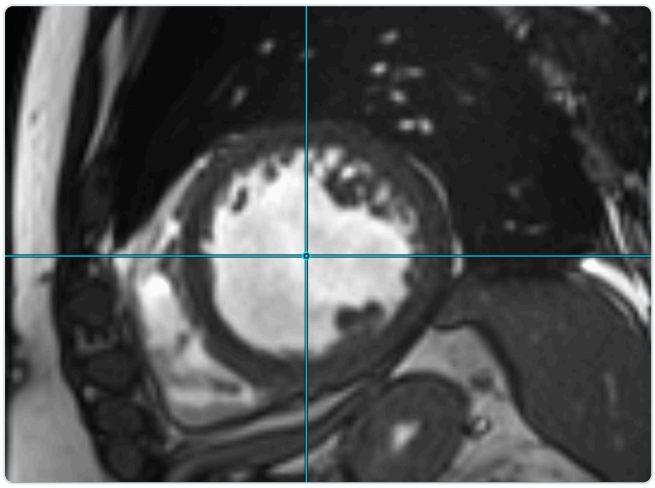
Inference device chooses CPU or a GPU. **Choose GPUs** is available for whole-cine segmentation and temporal propagation. Each selected device loads its own model the first time it is used.

> **Batch results are applied together after success.** Progress can advance before masks appear. **Cancel** preserves the annotation from before the job; one **Undo** restores a completed batch.

07 · PREDICT FROM YOUR CLICKS

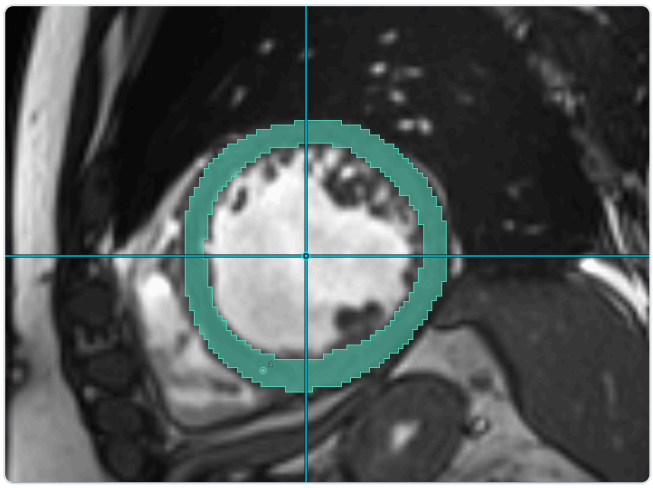
Start empty, then show the model what to include

Before · empty mask, Clicks only



No semantic prediction or imported mask is used as the starting annotation.

After · first positive point



On Frame 1, Layer 6 of a fresh study, the first point generates a new mask.

1. Use a fresh study, or reset a practice slice with **Clear all masks this slice**. This removes the working masks, prompts and prediction context at that position.
2. Select the organ that will receive the result, for example **MYO**.
3. Set **Click mode** → **Clicks only**. Do not run semantic initialization first for this exercise.
4. Choose **Include** and click inside the intended structure in the acquisition view. Wait for the new mask.
5. If needed, add another Include point to a missing region, or choose **Exclude** and click an unwanted region. Review each new result.

> **Selecting MYO chooses where the mask will be stored.** In **Clicks only** mode, the prediction uses point prompts without the organ-name query. To refine a semantic prediction with both inputs, use **Organ + clicks** (page 9).

Exclude can also be placed with Shift + click in Include mode. It asks the model to exclude that location; it is a new prediction, not a pixel eraser. Use Brush or Erase for exact manual changes.

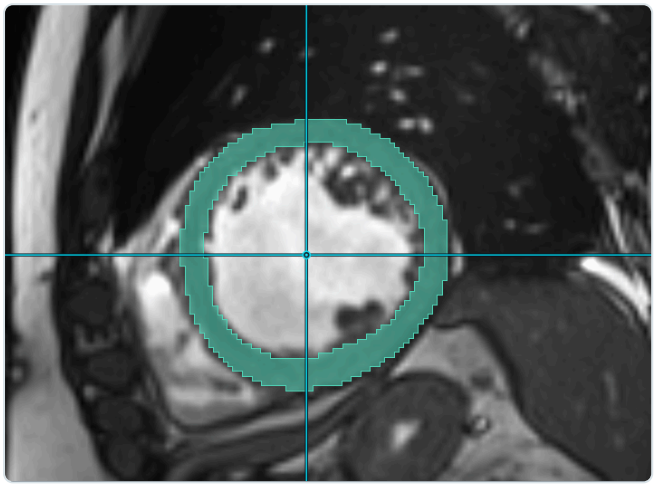
Clear operations change the working annotation, so practice in an example copy or save a backup first. Hiding a mask or its markers does not make the annotation empty. **Clear points** removes prompts but keeps the current mask. Changing Click mode starts a fresh prompt history with the next click.

08 · CORRECT A SEMANTIC PREDICTION

Start automatically, then refine with clicks

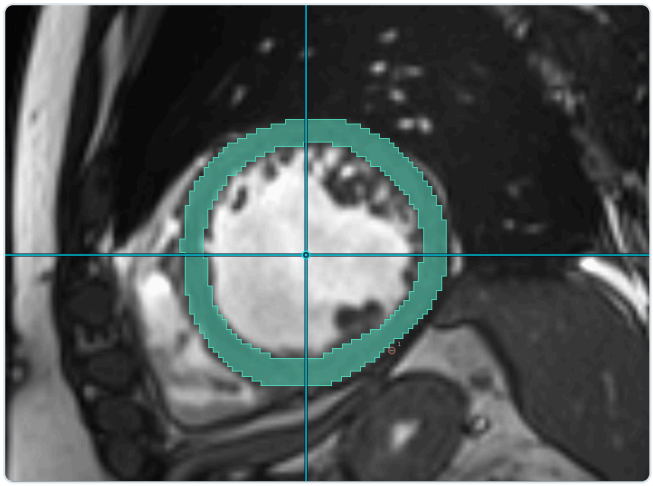
You can combine the organ-name prediction with your positive and negative points.

Before · semantic starting mask



Frame 1, Layer 6. A semantic MYO prediction supplies the initial mask and probability context.

After · Organ + clicks



One Exclude point near the lower-right boundary updates MYO using the organ name and preceding prediction.

1. Choose the organ, frame and layer. Run **Segment this slice**, or start from that organ's result after **Segment all**.
2. Keep the same organ selected and set **Click mode → Organ + clicks**.
3. Use **Include** inside a missed region, or **Exclude** inside an unwanted region.
4. Wait for the mask to update, then inspect the boundary. Add another point if useful.
5. Finish with **Brush** or **Erase** wherever an exact pixel correction is needed.

> **Choose the mode for your starting point. Organ + clicks** combines semantic and interactive guidance. Switching a semantic result to **Clicks only** starts a new prompt history and discards its semantic probability context; it is not the same refinement workflow.

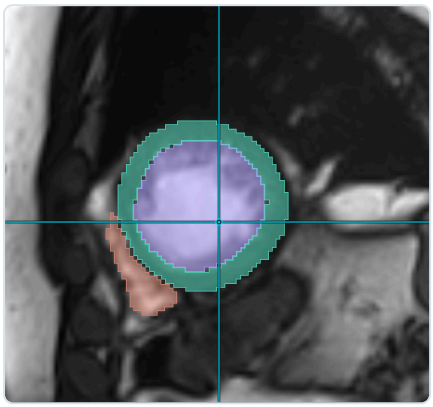
A point requests a new prediction, so it can change more than the pixel under the marker. Review the full visible boundary. Use **Point opacity** to see through markers without removing their effect.

The illustrated result is an actual model response, not a guarantee that every added point improves accuracy. Use Undo to revisit an earlier result, or manual tools to make a precise correction.

09 · PREDICT FROM A REFERENCE IMAGE AND MASK

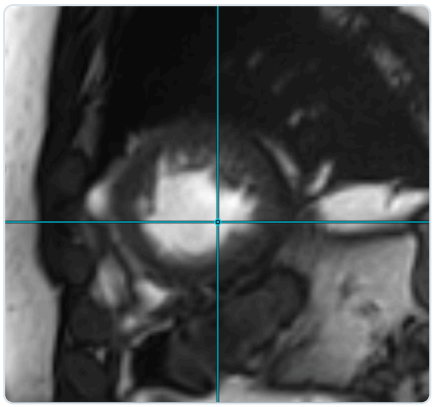
Guide a difficult frame with a good reference

1 · Frame 1 · reference



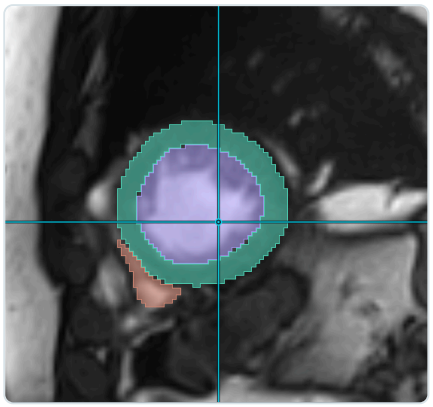
Layer 9: the image and its model-generated editable mask provide the example.

2 · Frame 16 · empty



The same Layer 9 at Frame 16 starts without a mask.

3 · Frame 16 · result



A real propagation job predicts a mask on the target image.

1. Pause on the layer of interest. Create and review a good editable mask in a reference time frame.
2. Expand **PROPAGATE IN TIME** under Edit History.
3. Choose **Use current frame**, or select **1–3 reference frames**.
4. Select the organs and inspect the target/skip summary. A target does not need a pre-existing mask.
5. Leave **Replace manual edits in target frames** off to retain manual corrections; locked targets are always protected.
6. Click **Propagate to N frames** and wait for completion. Navigate to a target frame and inspect its new mask.

> **A useful application: difficult basal and apical slices.** On that same acquisition layer, choose a time frame you can annotate reliably. Refine its mask, then use it as a reference for harder time frames. The model uses the reference image and mask as context; no retraining is performed.

The reference masks are frozen at job start. A **Reference comparison** overlay is only for viewing, not an input. If all selected references lack an organ, that organ is skipped.

Scope: the current layer across time; this control does not transfer a middle-layer mask to a basal or apical layer. References and locked targets are preserved. One **Undo** restores the completed job.

The illustrated near-apical example uses Layer 9: Frame 1 guides 29 other time frames, with an initially empty Frame 16 shown before and after. This demonstrates the workflow, not measured apical accuracy. Review every target.

10 · MANUAL REFINEMENT

Refine the result from any method



Check the organ and the tool

This detail shows **MYO** selected and **Erase** active. The size reads **1 pixel**: a tip one original image pixel wide.

Choose the tool from the toolbar, then use it in the acquisition view. The editing panel explains the current action.

The demonstration uses actual model predictions and illustrated edits. Clearing controls continue below the displayed detail; their scope is listed on page 16.

Control	How to use it
Include	Click inside a region that should belong to the selected organ. The model generates a new mask.
Exclude	Click an unwanted region. Shift + click in Include mode also adds a negative point.
Brush	Paint the selected organ directly. It takes priority over other unlocked organs at painted pixels.
Erase	Remove selected-organ pixels. The smallest brush or eraser tip is one original image pixel wide.
Undo / Redo	Reverse or restore an eligible action during the current server session.

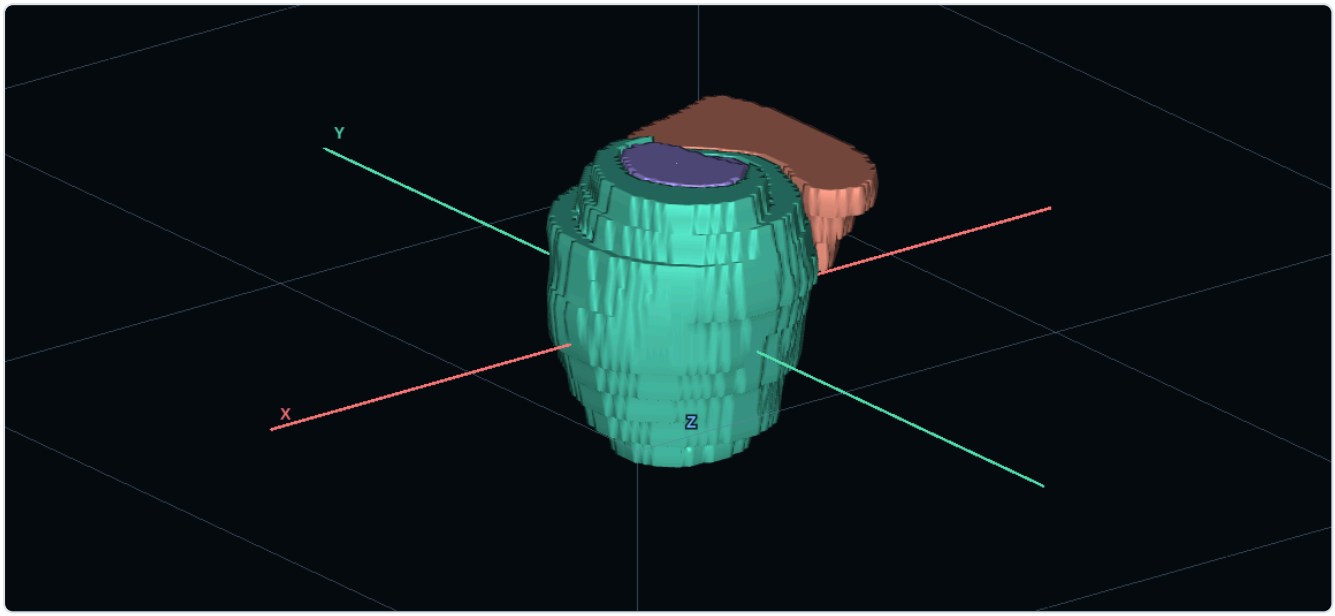
Zoom in for precise painting. Lower **Point opacity** if a marker hides a boundary. Lower **Mask opacity** to inspect the image underneath the mask.

To correct a semantic prediction, choose **Organ + clicks** (page 9). It retains the preceding semantic prediction as context. **Clicks only** is the separate exercise for building from spatial prompts. Wait for a prediction before judging it.

> **This is a 2D interaction model.** A click affects only the current time frame, acquisition layer and organ. It does not spread through the volume or movie. A new prediction may still need manual correction.

11 · REVIEW IN TIME AND SPACE

Play the cine; render 3D when paused



Current-frame model masks in an upright camera view. The camera is rotated for presentation; image and mask geometry are unchanged.

Play a 4D study

1. Use the CINE time slider or previous/next controls to choose a frame.
2. Click **Play**, or press Space.
3. Adjust FPS and looping if useful.
4. **Pause** before editing. Moving the time slider also pauses playback.

Inspect a 3D mask

1. Pause at the desired frame.
2. Click **Render 3D** in the locator view.
3. Drag in the 3D view to rotate it; adjust **3D mask opacity** or smoothing.
4. Choose **Hide 3D** to return to the locator alone.

> **Playback displays existing masks.** It does not create missing masks or copy a segmentation to another frame. Use whole-cine segmentation, temporal propagation or a matching 4D annotation to populate those frames.

For a full-volume surface, prepare masks across the relevant layers first, for example with **Segment all layers**. A mask on one slice can only produce a thin slice surface.

Playback, time changes and mask edits remove the surface; render again when needed. Smoothing affects appearance only, never mask pixels.

This public ACDC cine has no encoded patient orientation. Its image axes should not be read as confirmed anatomical directions. Camera rotation changes the viewpoint only. FPS is a display target, not a heart-rate measurement.

12 · FRAME × LAYER LOCKS

Protect exactly the slices you choose



Detail: Frames 1–2 × Layers 5–6 protects four slices. The “2/10” badges mean two of each frame's ten layers are locked.

1. Open **Edit History**. Use **Lock slice** for the current frame/layer.
2. For a range, choose **Select / Lock**.
3. Select numbers in both **FRAMES** and **LAYERS**. Shift + click selects a continuous range.
4. Check the displayed scope, then click **Lock selected**.
5. Use **Done selecting** to return to normal navigation. To edit those slices later, select them and choose **Unlock selected**.

> **Example:** Frames 2–4 × Layers 3 and 5 protects **six slices**, including all three organs at each position. **All layers** deliberately selects entire volumes for the chosen frames; **All frames** selects the chosen layers throughout the cine.

Locks protect masks, clicks and refinement context from segmentation, propagation, imports, clearing and editing. **Replace manual edits** cannot override them. Undo/redo cannot change a newly locked slice either.

Empty slices can be locked. Locked slices remain available for viewing, exporting and propagation references. The separate lock beside an organ protects that organ across the study.

Lock metadata saves separately; use **Save study** for any pending annotation edits. A partially locked frame reports the number of protected layers.

13 · HISTORY, NOTES AND COMPLETION

Leave a clear place to resume



Notes apply to the whole study. Completion is a workflow marker that can be reversed with Reopen.

Understand Edit History

Empty, **Mask present** and **Edited** distinguish annotation states. A lock icon marks protection. Collapse the panel with its arrow to make more room.

Activity log lists operations in words. Switch between **Current slice** and **Whole study**; use **Load older activity** for earlier records.

Write a useful note

In **STUDY NOTE**, record what was reviewed or what needs another look. For example: “MYO reviewed on frames 1–8, layers 3–7. Recheck basal boundary in frame 9.”

Wait for **Note saved** or click **Save note**. Notes save separately from pending annotation pixels.

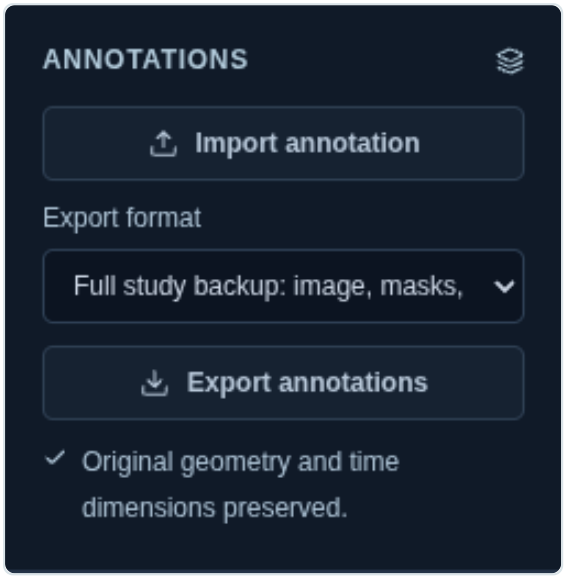
Annotation complete or **Complete** saves the annotation and note before marking the study complete. Find it under **Completed** in the Library. **Reopen** returns it to work in progress.

> **Complete means “marked complete by a user.”** It does not certify mask accuracy, prove that every frame was segmented, or replace slice locks.

The activity log persists with the study. Undo/redo is a separate session history and is cleared when the server restarts.

14 · KEEP YOUR WORK

Save on the server; download a portable backup



Save and export have different destinations

Save study, in the top bar, preserves working state on this server.

In the annotation panel, select **Full study backup** and click **Export annotations** to download a portable copy.

Your browser may ask where to save it; otherwise, check its Downloads folder.

Choose	For this purpose
Save study	Continue later on this server. Check that **Unsaved edits** clears.
Full study backup · ZIP	Keep or transfer the complete image, masks, reference, notes, locks, prompts, activity and refinement context.
Combined labelmap · NIfTI	Use one segmentation volume elsewhere: 0 background, 1 RV, 2 MYO, 3 LV.
Separate masks + edit record · ZIP	Use individual organ masks and an annotation record elsewhere. This is not a full-study restore archive.
RV / MYO / LV binary mask	Download one structure as a NIfTI mask.

Choose an **Export format**, then **Export annotations**. Exports preserve the source geometry and time dimensions. A combined labelmap requires nonoverlapping masks; use **Resolve · prioritize [organ]** or separate-mask export if overlap remains.

Restore a full study: return to Library and open its ZIP through **Open your NIfTI volume**. Use **Import annotation** only for aligned mask NIfTI files.

> **Undo/redo does not survive restart or backup import.** A saved study or full backup keeps the current annotation state, rather than the session's undo stack.

Delete in My studies removes that server-side study. Back it up first if it may be needed. Removing an example hides its Library entry; permanently removing that entry does not delete separately saved studies.

15 · AT A GLANCE

The remaining controls

Navigation and display	Effect
Library	Open examples, saved studies, completed studies or a new input.
Image geometry	Inspect dimensions, spacing, preprocessing and input warnings.
Eye icon / Reference contours	Show or hide an organ or comparison overlay; annotations stay unchanged.
Contours only / Mask opacity	Choose outlines or change the 2D overlay's transparency.
Locator color / opacity	Adjust the shared crosshair and 3D locator.
Point opacity / Hide / Show	Change click-marker visibility; the prompts remain active.
Window / Level / Reset display	Adjust image brightness and contrast, or restore the initial display.
Render 3D / Hide 3D	Build current-frame mask surfaces, or release them.
3D mask opacity / Smooth 3D surfaces	Change surface appearance without changing mask pixels.

Clearing and protection	Scope
Clear points	Selected organ's prompts on the current frame/layer.
Clear this slice	Selected organ's annotation on the current frame/layer.
Clear all masks this slice	RV, MYO and LV on the current frame/layer.
Clear all masks this layer	RV, MYO and LV at the current layer across unlocked frames. For a 3D image, this is one slice.
Resolve · prioritize [organ]	Resolve current-slice overlaps in favor of the selected organ, respecting protected structures.
Organ lock	Protect one organ throughout the study.
Lock slice / Lock selected	Protect all organs at the selected frame/layer positions.
All frames / All layers / Clear selection	Change the lock selection scope; these controls do not themselves change mask pixels.

All-mask clearing requires organ locks to allow it. Slice locks remain protected. A configured library may also offer **Load matched annotation** or **Load evaluated 3D example**; these load supplied masks rather than running a new prediction.

> **A disabled control usually has a reason:** playback is active, a task is running, the study is complete, or a slice/organ is locked. Pause, wait, reopen or unlock as appropriate.

16 · WHEN YOU NEED A REMINDER

Common questions

What you notice	What to check
The browser cannot connect.	Is the service running? Is the SSH terminal open? Does the browser port match the first port in `-L`?
A frame has no mask.	Playback displays existing masks only. Segment, propagate or import that position.
A large job has no visible result yet.	Check preparation and per-GPU progress. Results are committed together after success.
The movie is slower than requested.	Allow the initial frame load. Use browser hardware acceleration and try a lower FPS. Server inference and browser rendering use different GPUs.
The 3D mask disappears.	This is expected after playback, time changes or editing. Pause and choose Render 3D again.
A mask will not import.	Check the full spatial geometry and time dimensions against its source image.
A correction cannot be made.	Pause playback, wait for jobs, inspect slice/organ locks, and reopen a completed study if needed.

Keyboard shortcuts

1 / **2** / **3**	Include / Exclude / Locate	4 / 5	Brush / Erase
6 / **7** / **8**	Pan / Zoom / Window	↑ / ↓	Previous / next layer
R	Fit views	Space	Play / Pause
Ctrl/⌘ + S	Save study	Ctrl/⌘ + Z	Undo
Ctrl/⌘ + Shift + Z	Redo; Ctrl+Y also works	Esc	Pause / close dialog

Shortcuts apply when focus is outside a text field or menu.

More detail and sample attribution

User guide · Installation · Model card · Example data. Repository links require access while the project is private.

Screenshots use the public Automated Cardiac Diagnosis Challenge (ACDC) example. Its data have separate **CC BY-NC-SA 4.0** terms for noncommercial scientific research. The 4D image has 216 × 256 × 10 × 30 voxels; the separate expert annotation covers the end-diastolic phase, not all 30 time frames. Refer to each workflow caption for the origin of the displayed masks.

O. Bernard, A. Lalande, C. Zotti, F. Cervenansky, et al. “Deep Learning Techniques for Automatic MRI Cardiac Multi-structures Segmentation and Diagnosis: Is the Problem Solved?” *IEEE Transactions on Medical Imaging* 37(11), 2514–2525 (2018). doi:10.1109/TMI.2018.2837502.

The workstation and released model use PolyForm Noncommercial 1.0.0; upstream components retain their own licenses. For a support report, include the version, browser, device and a de-identified description of the problem.